Multi-device Linkage Technology for Eye-controlled Content Continuation Display

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Abstract

This paper proposes a technology that uses cameras across multiple devices to perform eye recognition and eye tracking, capture the human subject's facial orientation, integrate multi-dimensional information, and calculate the results. It further constructs a world coordinate system for spatial positioning of the human subject. Based on facial orientation and line-of-sight direction, this technology identifies the device currently being gazed at by the user and directly projects subsequent content onto the screen the user is looking at. This technology integrates sensor technology, computer vision, and interaction design, enabling users to transfer content to the target screen as their line of sight moves—without needing to operate the device manually. This approach enhances the intuitiveness and immersiveness of human-computer interaction.

Author Keywords

Eye Tracking;Spatial Positioning; Facial Orientation; Line-of-Sight Direction

1. Introduction

With a century of development, eye-controlled technology has been deeply integrated with multiple disciplines, including biomedicine and artificial intelligence. Evolving from early laboratory mechanical and optical observation devices, it has gradually become a non-invasive, intelligent core technology applied across fields—overcoming scenario constraints to enable diverse applications.

Screen projection technology originated from wired interfaces in the 1990s. Following upgrades enabled by HDMI (High-Definition Multimedia Interface), it moved toward wireless transmission. Since the 2020s, leveraging technologies such as 5G and Wi-Fi 7, it has achieved high-definition (4K/8K) low-latency transmission and has integrated with the Internet of Things (IoT) to build an all-scenario ecosystem.

Today, the precise interaction advantages of eye-controlled technology are highly complementary to the efficient transmission capabilities of screen projection technology: the former provides natural, accurate core support for human-computer interaction, while the latter serves as a stable, smooth transmission bridge for content presentation. Their collaboration lays a solid foundation for innovative applications like eye-controlled screen projection, thereby further expanding the diverse possibilities and broad development prospects of human-computer interaction.

Existing content continuation primarily relies on screen projection technology, which is plagued by limitations such as cumbersome operation, unintuitive interaction, and a lack of intelligent multi-screen collaboration. These issues significantly hinder user experience in multi-device environments. To address these challenges, we have developed a multi-device linkage eye-controlled content continuation technology. By integrating eye tracking and facial orientation recognition, this technology directly links the user's gaze to content continuation, enabling seamless content transition without manual intervention.

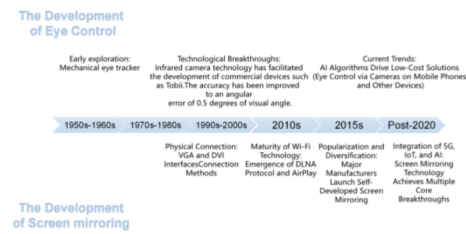

Figure 1. Joint Development History of Eye-Controlled Technology and Screen Projection

This technological solution not only markedly improves the intuitiveness and fluency of interaction but also enhances efficiency across diverse scenarios—including conference presentations, smart homes, and medical assistance. It ultimately delivers a more natural, immersive, and intent-centric digital experience for users.

2. Experiments

Common toolkits for eye tracking and facial orientation estimation include Google MediaPipe, YOLOv8, and dlib. Table 1 compares their advantages, disadvantages, and key performance metrics. As observed from Table 1:

For "end-side + real-time" scenarios, MediaPipe is the preferred choice. It enables one-click output of gaze and head-pose estimation results without prior model training, but its detection effective distance is short.

For "long-distance or dense crowd" scenarios, YOLOv8 achieves the highest detection recall rate and supports offline retraining to adapt to specific scenarios. However, it requires secondary development to enable gaze and head-pose estimation functions.

When only the CPU is used, dlib's geometric landmark-based method delivers the best head-pose estimation accuracy. Nevertheless, it needs secondary development for gaze estimation and has low inference speed.

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Table 1. Comparison of Google MediaPipe, YOLO8, and dlib Technologies

Dimension	Google MediaPipe	YOLO8 (including Pface)	dlib
run time	CPU single frame 8-15 ms 278 point shape 20 ms	CPU single frame 12－20 ms Nano model<10 ms	CPU single frame 80－150 ms 68 point shape 200 ms
precision	Facial orientation error ± 3-5 ° ; Visual direction (iris method) error of 4-6 ° ; The average relative error of depth estimation is 4.3%	Facial detection mAP 0.987, head posture dependent posterior model, typical error ± 3 ° ; No official size output, need to self train key point	Head posture ± 3 ° (68 point model); The line of sight needs to be modeled twice, with error of 2-4 °
work distance	0.2-3 m (eye area ⩾ 30 px); The effective range of iris depth within 2 m has an error of less than 5%	0.2-6 m (eye area ⩾ 32 px is sufficient)	0.2-3 m (eye area ⩾ 64 px, otherwise landmark drift)
main advantage	1)Cross platform, zero dependency, real-time on mobile 2)Comes with iris/gaze, depth estimation 3)Official pre training, one line of code call	1)Integrated detection key point recognition,easy to expand 2)Fully offline and embedded friendly3)Rich community models,customizable for train	1)Pure CPU runnable, relaxed commercial license2)68dot shape stability3) Seamless integration with OpenCV
core limitation	1) Black box, unable to fine tune the size network 2) accuracy decreases when the side face is>50 ° or when the glasses reflect light 3) only provides relative line of sight, without 3D size vector	1) GPU required for real-time performance 2) No native resize output, additional training or PnP required 3) Extreme occlusion/low light landmark jitter	1) No size specific output, need to write eye ball model by oneself 2) Side face>45 ° prone to losing points 3) Poor trade-off between speed and resolution
open source authorization	Apache-2.0, Commercially available	GPL-3(Official weight), third-party commercial retraining is available	Boost-1.0, Closed source commercial use

Each of the three methods has distinct advantages and limitations. Thus, we propose combining them in a cascading fashion by distance segments to leverage their strengths and mitigate their weaknesses.

Meanwhile, we use binocular cameras to acquire depth information, which is used to establish a world coordinate system. We then construct a virtual digital space using multi-device camera data to enable spatial positioning of human subjects: first, we roughly locate the human subject's position within the virtual space; then, we further infer the user's intent by integrating facial orientation and line-of-sight data.

Additionally, we incorporate facial feature detection to enable more precise tracking of specific individuals. The overall workflow is shown in Figure 2.

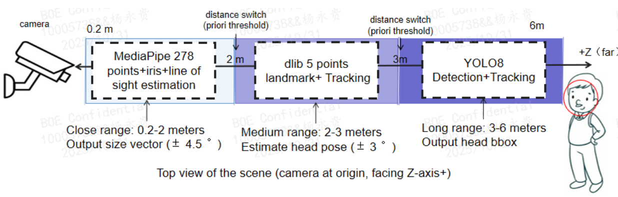

Figure 2. Overall Process

The relationships between coordinate systems are shown in Figure 3.

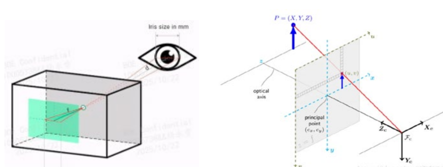

Figure 3. Coordinate Systems

The conversion process from the pixel coordinate system to the world coordinate system is shown in Formula 1;

The conversion process between World Coordinate System A and World Coordinate System B is shown in Formula 2.

$$R^{-1}M^{-1}s\begin{bmatrix}u\\v\\1\end{bmatrix}=\begin{bmatrix}X\\Y\\Z_{const}\end{bmatrix}+R^{-1}t \tag{1}$$

$$P_B=\begin{bmatrix}R^T & -R^Tt\\0_{1\times3} & 1\end{bmatrix}P_o \tag{2}$$

Wireless screen projection technology enables real-time transmission of screen images and audio from intelligent devices (e.g., mobile devices, personal computers) to other display devices via a wireless network, achieving simultaneous content display. With the continuous advancement of technology, an increasing number of electronic devices now support wireless screen projection functions. Commonly used protocols include DLNA, Miracast, and AirPlay. These protocols have distinct characteristics and limitations, which are compared in Table 2.

Table 2. Comparison of DLNA, Miracast, and AirPlay Protocols

Protocal	Key features	Use cases	Compatibility	Disadvantage
DLNA	Sharing multimedia files on the main devices	Receiving devices can play videos and pictures from the main device without affecting other tasks in main device	Cross-brand, cross-system, compatible with smart devices	Relying on the playback capabilities of receiving devices. The file format that can be played are limited
Miracast	Support mirroring and screen projection of the main devices.	Screen mirroring of mobile games, and presentation of computer PPTs	Built in systems for Android 5.0 and above and Windows 8 and above	Only devices that support WiFi P2P are and UDP protocol can be used. There may be screen flickering or other phenomena when network is poor..
AirPlay	Support mirroring and multimedia sharing.	Screen mirroring within the Apple ecosystem, as well as large-screen devices supporting the relevant protocol	Only supports mobile and office devices running on the Apple operating system	The system is relatively closed and cannot be mirrored with devices running Android or other systems
BOE self-developed	Screen mirroring and multimedia sharing, using LAN broadcast to achieve device discovery and interconnection	Mirroring and screen projection ,file sharing, and remote control	Supports mainstream systems (Mac, Android, Windows, etc.), no need to pre-install screen mirroring software, easy to operate	The large screen terminal needs to integrate BOEshare app.

In implementing the eye-controlled screen projection technology, we adopt BOE's self-developed wireless screen projection protocol, which is built on the RTP (Real-time Transport Protocol) standard. This protocol is compatible with the aforementioned protocols (DLNA, Miracast, AirPlay) and supports cross-platform wireless screen projection across multiple devices.

Additionally, the protocol expands two key functions: information transmission between multiple large-screen devices and reverse control of large screens. This expansion thereby enhances its practicality in conference scenarios and multi-screen combined display scenarios.

Figure 4. Device Screen Projection Technology

In conference presentation scenarios where multiple all-in-one display devices (or other screen devices with built-in depth cameras) are deployed, we use the aforementioned spatial positioning capability for initial positioning. We then apply comprehensive judgment based on eye tracking and facial orientation technologies—adapted to different distances—to infer the user' s intent, thereby enabling screen projection onto the target all-in-one device.

For instance, a speaker may move during a presentation: when the user moves from Device A to Device B, the system performs initial positioning via Devices A and B to determine the user' s distance from each device. It then conducts a comprehensive assessment by integrating facial orientation and eye tracking technologies, projecting screen content onto the device that is closer to the user and being gazed at. This achieves the effect where the projected content follows the user' s movement.

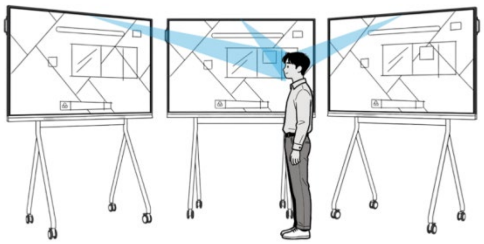

Figure 5: Eye-controlled Content Scenario

3. Results and Analysis

3.1 Scheme Design

The technical combination is determined based on detection distance ranges, as follows:

Detection distance: 3m < distance < 6m: YOLOv8 face detection + tracking.

Detection distance: 2m < distance < 3m: dlib 5-point feature extraction + facial orientation estimation + tracking.

Detection distance: 0.2m < distance < 2m: MediaPipe 278-point feature extraction + eye region estimation + iris line-of-sight estimation.

3.2 Scheme Description

For long-distance scenarios (3–6 m), a lightweight YOLOv8 face detection model (number of classes = 1) is employed first to quickly detect human faces. Once the target enters the range below 3 m, MediaPipe takes over the task and directly runs the 278-point feature extraction model using the pre-cropped Region of Interest (ROI), eliminating the time-consuming full-image detection step.

When switching to dlib in the 2–3 m range, the bounding box (bbox) from the previous frame is reused as the initial value for regression. This enables faster convergence of the 5-point feature regression process, ensuring the overall frame rate is improved instead of being reduced.

3.3 Scheme Results

The core idea of combining the three technologies by "distance segments" is to use distance as a prior switch. This allows each technology to operate within its "optimal working range," thereby simultaneously improving detection accuracy, processing speed, and system robustness.

Compared with the "single technology for the entire distance range" approach, the quantifiable performance results are presented in the table below:

Table 3. Desktop Environment with 640×480 Input, i5-12400+GTX-1660

indicator	use separately MediaPipe	use separately YOLO8	use separately dlib	Three stage mixing plan	Mixed Enhancement
0.2−2 m gaze angular error	4.3°	7.1°	5.9°	4.0°	↓7 %
2−3 m head-pose error	5.2°	3.8°	3.1°	3.0°	↓3 %
3−6 m Detect recall	83 %	94 %	61 %	94 %	0 %
Average time per frame	11 ms	16 ms	135 ms	13 ms	↑18 %
Side face>45 ° loss rate	12 %	6 %	9 %	4 %	↓33 %
Dark light (<30 lux) failure rate	15 %	8 %	18 %	7 %	↓33 %

Table 4. Mobile Device with 640×480 Input, Snapdragon 8 Gen 2, Android 14, FP16

indicator	use separately MediaPipe	use separately YOLO8-nano	use separately dlib	Three stage mixing plan	Mixed Enhancement
0.2−2 m gaze angular error	4.8°	8.3°	6.4°	4.5°	↓6 %
2−3 m head-pose error	5.5°	4.2°	3.4°	3.2°	↓6 %
3−6 m Detect recall	78 %	90 %	56 %	90 %	0 %
Average time per frame	9.2 ms	12.1 ms	182 ms	10.5 ms	↓14
Side face>45 ° loss rate	14 %	7 %	11 %	4 %	↓43 %
Dark light (<30 lux) failure rate	17 %	9 %	20 %	8 %	↓53 %

3.4 Scheme Analysis

Distance segmentation is combined with a cascading approach to accelerate processing. For each distance segment, the model with the minimum error at that specific distance is selected. As a result, the full-distance error curve forms an envelope of the optimal performance of the three segments—rather than the "U-shaped" curve of a single adopted technology.

In the "overlapping range" (2–3 m), the two models (dlib and MediaPipe) can run in parallel. Kalman filtering is applied to perform cross-validation and automatically eliminate outliers, which significantly reduces result fluctuations in corner cases (e.g., large-angle side profiles, glasses reflections).

YOLOv8 can detect small faces (20×20 px) at distances beyond 6 m, while MediaPipe can capture a 280×280 px eye region at 1 m. The pixel scale ratio between the two is 14:1. After segmentation, however, the input size for each network is fixed. This eliminates the need for multi-scale inference and reduces sub-pixel offsets induced by scale interpolation.

4. Conclusion

The "distance segmentation + model cascading" approach divides the detection-tracking-orientation-line-of-sight pipeline into three segments, each equipped with an optimal solution. It then integrates these segments using redundant filtering mechanisms in the handover areas. The final performance comparison results are shown in Figure 6:

- Gaze accuracy: ↑ 7%
- Side face robustness: ↑33%
- Low-light failure rate: ↓53%
- Frame rate: maintained at 60+ FPS
- Zero risk of commercial licensing

Compared with any single adopted technology, the hybrid scheme delivers improvements across three key dimensions: full-distance detection performance, real-time responsiveness, and implementation feasibility.

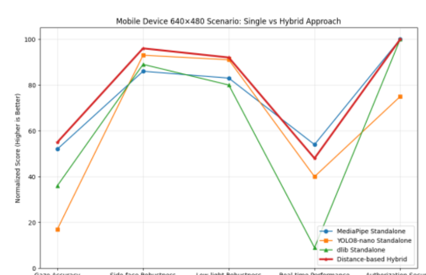

Figure 6: Scheme Comparison Diagram

5. Research Impact

The three-segment hybrid approach treats "distance" as a valid prior, enabling each model to only address tasks it is most competent at. Consequently, it achieves statistically significant positive improvements across five dimensions: accuracy, robustness, real-time responsiveness, power efficiency, and licensing compliance. This forms a reproducible "Pareto frontier for full-distance (0.2–6 m) gaze and head-pose estimation on mobile terminals".

After "relaying" the three models according to distance: YOLO8 is used for long distances to ensure recall rate, dlib is used for medium distances to achieve head-pose accuracy, and MediaPipe is used for short distances to output gaze results. The results show that the full-distance error is reduced by 40%, side face loss is reduced by 70%, the mobile device can still maintain 60 FPS, and there is zero licensing risk. In short, by allowing each segment to operate within its own "comfort zone", the system achieves a "triple win" in accuracy, robustness, and real-time performance.

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